

The Governing Heat Equation

For a one-dimensional rod, the heat conduction is governed by:

$$\rho c_p \frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial x^2}$$

- $\frac{\partial T}{\partial t}$ = the rate of change of temperature with respect to time ($^{\circ}C/s$)
- $\frac{\partial^2 T}{\partial x^2}$ = sensitivity of temperature changes at a specific point from its neighbors ($^{\circ}C/m^2$)
- t = Time (s), • x = Position along the rod (m), • $\alpha = \frac{k}{\rho c_p}$ is the thermal diffusivity (m^2/s)
- ρ = Density (kg/m^3) • k = Thermal Conductivity ($W/m \cdot ^{\circ}C$), • c_p = Specific Heat ($J/kg \cdot ^{\circ}C$)

Boundary and Initial Conditions

The model uses the Dirichlet boundary conditions

$$\begin{aligned} T(0, t) &= 100^{\circ}F && \text{(Left Boundary)} \\ T(L, t) &= 20^{\circ}F && \text{(Right Boundary)} \\ T(x, 0) &= T(x) && \text{(Initial Temperature)} \end{aligned}$$

Finite Element Analysis

The rod is divided into n elements of length $h = L/n$.
sFor each element, the local stiffness matrix $\mathbf{K}^{(e)}$
and force vector $\mathbf{F}^{(e)}$ are defined as:

1. Spatial Discretization (FEM):

The rod is divided into n elements of length h .
The local connectivity is defined by:

$$\mathbf{K}^{(e)} = \frac{k}{h} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

2. Transient System (Backward Euler):

To ensure unconditional stability, an implicit scheme is used:
 $(\mathbf{I} - \alpha \Delta t \mathbf{K}') \mathbf{T}^{n+1} = \mathbf{T}^n + \mathbf{b}_{bc}$

3. Boundary Treatment (Dirichlet):

The vector $\mathbf{b}_{bc}$ enforces the fixed temperatures at the boundaries ($100^{\circ}F, 20^{\circ}F$),
effectively driving the heat flow toward a linear steady-state where $\mathbf{K} \mathbf{T} = \mathbf{0}$.

Accounting for ambient air influence, the 1D heat conduction is governed by:

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} - \nu(T - T_\infty)$$

- ν : Convection Coefficient (s^{-1})
- T = Temperature ($^{\circ}\text{C}$),
- T_∞ = Ambient Temperature ($^{\circ}\text{C}$)

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Property	Copper (Bulk)	Additive (3D-Printed)	Physical Interpretation
Conductivity (k)	384.72 W/m·K	14.40 W/m·K	XN is $\approx$ 26x more resistive to heat flow than Cu.
Diffusivity (α)	0.0738 m ² /s	0.0213 m ² /s	Cu reaches steady state $\approx$ 3.5x faster than XN.
Convection (h)	165.41 W/m ² ·K	128.70 W/m ² ·K	Variations due to ambient lab/wind conditions.
Avg. Std. Error	0.95	0.73	Low error validates 1D FEA model robustness.

Table 1: Thermal Parameter Comparison: Bulk Copper vs. 3D-Printed Steel (XN)

Property	Copper	3D Printed	Physical Interpretation
Conductivity (k)	384.72 W/m·°C	14.40 W/m·°C	3DP is $\approx$ 26x more resistive to heat flow than Cu.
Diffusivity (α)	0.0738 m ² /s	0.0213 m ² /s	Cu reaches steady state $\approx$ 3.5x faster than 3DP.
Convection (h)	165.41 W/m ² · ° C	128.70 W/m ² · ° C	Variations due to ambient lab/wind conditions.
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